Involuntary Part-Time Employment, Gig Rollout, and Flexibility of Work

Matt Eaton*

John E. Walker Department of Economics, Clemson University, 320 Wilbur O. and Ann Powers Hall, Clemson, 29634, SC, USA

Srijan Banerjee

Francis Marion University, School of Business, 4822 E. Palmetto St., Florence, 29506, SC, USA

Abstract

The Great Recession saw not only high unemployment, but also a sharp rise in involuntary part-time employment—workers seeking full-time work but finding only limited part-time hours. The involuntary part-time employment rate declined gradually throughout the mid-to-late 2010s, a period that coincided with the rapid expansion of gig employment. This paper examines whether the introduction of UberX, a major ride-hailing platform, causally reduced involuntary part-time work in metropolitan area labor forces. We utilize the geographic and temporal rollout of UberX across metropolitan areas to estimate causal effects via a staggered Difference-in-Differences strategy. Our results indicate that UberX entry reduces involuntary part-time employment as a share of the total labor force by 10 percent and as a fraction of total part-time employment by 8.7 percent. We find little evidence of other labor market flows during this period, suggesting that UberX serves as a mechanism for workers to transition from involuntary to voluntary part-time employment.

Keywords: Involuntary part-time employment, Gig economy, Ride-hailing, Underemployment, Labor market flexibility

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*Corresponding author. Email addresses: mreaton@clemson.edu (Matt Eaton), srijan.banerjee@fmarion.edu (Srijan Banerjee)

1. Introduction

According to the Bureau of Labor Statistics (BLS), a worker is classified as “involuntary part-time employed" if they work in a part-time position (typically fewer than 35 hours per week) but would prefer a full-time position and are unable to secure one due to economic factors, such as inadequate labor demand. Speaking from an allocation standpoint, the presence of involuntary part-time employment (IPE) reflects the under-utilization of available labor resources and is a critical measure of labor market slack (Blanchflower and Levin [1]; Borowczyk-Martins and Lalé [2]). Since the 1970s, the involuntary part-time employment as a fraction of the labor force in the US has been stable at around 2.0 to 4.0 percent, with fluctuations occurring during recessionary periods, giving involuntary part-time employment a counter-cyclical nature similar to unemployment. Involuntary part-time employment as a fraction of the total labor force reached a historical high during the Great Recession when it rose to 6.0 percent, and experienced a gradual return to the historical trend during the mid-2010s.

During a similar time frame of recovery, the US labor market experienced a significant rise in the prevalence of platform-based independent contract work, commonly referred to as “Gig" employment. Such employment opportunities have increased the flexibility that workers enjoy, both at the extensive and intensive margins. This can be attributed to factors such as higher flexibility in working hours, location of operation, and lower entry costs of employment. Since the emergence of gig work, there has been a sharp rise in platform-based gig work participation, driven mainly by workers in traditional jobs seeking supplemental income (Collins et al. [3], Hall and Krueger [4], Farrell et al. [5]). While numerous studies focus on the impact of gig employment on labor market outcomes, such as unemployment, earnings, entrepreneurial activity, and many others, its relationship with involuntary part-time employment remains largely unexplored (Huang et al. [6],Burtch et al. [7],Chen et al. [8],Garin et al. [9],Collins et al. [3],Jackson [10]).

In this paper, we investigate the effect of introducing gig employment opportunities on involuntary part-time employment (IPE) in the United States. To this end, we exploit the quasi-random nature of the rollout of UberX services across metropolitan areas (henceforth referred to as metareas) in the United States between 2010 and 2019. "From the Current Population Survey (CPS) microdata, we construct metropolitan area-level variables including involuntary part-time employment as a fraction of the labor force and as a fraction of total part-time employment. For the data on the rollout of UberX services, we follow Hall et al. [11], which provides the official entry date of the service in an area based on UberX blog posts announcing the service. Using a staggered Difference-in-Difference (DiD) framework *a la* Callaway and Sant'Anna [12], we provide causal estimates of the effects of exposure to UberX services at the metarea level on involuntary part-time employment as a share of total labor force as well as the involuntary part-time employment as a share of total part-time employment.

Our results indicate that the advent of platform-based gig work opportunities, such as UberX, played a crucial role in shaping the decrease in involuntary part-time employment. We find that metareas exposed to UberX services experienced a 10 percent drop in involuntary part-time employment as a share of the labor force. On the other hand, as a fraction of total part-time employment, there was an 8.7 percent decline in involuntary part-time employment for metareas exposed to UberX services. To put these estimates into perspective, we examine the impact of UberX's rollout on other labor market variables, including the unemployment share, the share of the labor force working full-time, and the share of the labor force working part-time. We find no significant treatment effects on any of these variables. This leads us to conclude that the advent of UberX can be viewed as a mechanism for worker transition from involuntary to voluntary employment. We hypothesize that this worker reallocation takes place primarily because of the flexibility of hours and location of operation offered by gig work, such as UberX.

Examining heterogeneity across the adoption group reveals that metropolitan areas exposed to UberX in 2013, 2014, and 2016 exhibit significant declines in involuntary part-time employment, while metropolitan areas that received treatment in 2015 and 2016 show no significant response. Notably, the 2014 adoption group represents the largest one-year expansion of UberX across U.S. metropolitan areas in our sample, making it the primary driver of our main results. Importantly, the timing of these effects varies: dynamic analysis shows that the 2013 and 2014 adoption groups exhibit delayed responses, with significant effects emerging one to two years post-treatment, while the 2016 group shows immediate impacts. These patterns converge around 2016-2017, coinciding with the surge in UberX adoption and infrastructure.[1] We argue that early adopters (2013-2014) required time to build infrastructure and disseminate knowledge of the UberX service before impacts materialized, while by 2016, Uber's rapid expansion enabled immediate labor market effects. The lack of significant effects for the 2017 group may reflect reduced experimentation as the costs and benefits of gig work became clearer, or alternatively, that effects were concentrated in areas with established infrastructure during the 2016 surge.

Our paper contributes to two major strands of literature. First, it complements a growing body of research that examines the impact of expanding opportunities in gig employment on labor market outcomes (see Abraham et al. [14]; Abraham et al. [15]; Abraham et al. [16]; Angrist et al. [17]; Berger et al. [18]; Hall and Krueger [4];Hall et al. [11];Cramer and Krueger [19]; Farrell et al. [5]; Chen and Sheldon [20]; Chen et al. [8]). As highlighted in Katz and Krueger [21], there has been a rise in the incidence of alternative working arrangements in recent years, attributing the change to potential factors such as access to new technology that facilitates the assignment of contract work and a growing demand for more flexible work arrangements.[2] They also acknowledge the role of slack labor markets and higher unemployment in the increasing incidence of alternative work arrangements, but find that these factors play only a limited role in driving the transition. Our results provide support by highlighting the reallocation of workers within part-time employment as a response to exposure to gig employment opportunities during a time when the economy was recovering from the Great Recession.

On the other hand, this paper extends the body of work that focuses on involuntary part-time employment in the United States (see Borowczyk-Martins and Lalé [23]; Valletta et al. [24]; Borowczyk-Martins and Lalé [25]; Cajner et al. [26]; Kudlyak et al. 27; Glauber [28]; Glauber [29]; Stratton [30]; Borowczyk-Martins and Lalé [31];Golden and Kim [32];Canon et al. [33];

[1]According to internal Uber company data reported by Iqbal [13], the amount of Uber capital funding received by the company grew from 0.44 billion in 2013 to 4.64 billion in 2014 to 9.14 billion in 2016. The number of Uber users increased from 11 million in 2015 to 37 million in 2016 and further to 58 million in 2017. During the same period, revenue from gross bookings surged from $ 19 billion in 2016 to $ 45 billion in 2017. Finally, the number of Uber trips increased from 3.79 billion in 2017 to 5.21 billion in 2018.

[2]See also Katz and Krueger [22] for a detailed discussion on increase in alternative work arrangements as measured using administrative data. They define alternative work arrangements as temporary help agency workers, contract workers, on-call workers and independent contractors or freelancers.

Larson and Ong [34]). Our paper is closely related to Borowczyk-Martins and Lalé [23], which highlights the counter-cyclical nature of involuntary part-time work and analyzes worker flows between involuntary part-time employment and other employment states. As pointed out by Kudlyak et al. [27], the increase in involuntary part-time employment during the Great Recession was primarily driven by transitions from full-time employment and from voluntary part-time employment. However, during the recovery period, the transition rate from involuntary part-time to full-time employment remained below its pre-recession level, indicating persistent challenges in securing full-time work. There were considerable increases in worker flows from involuntary to voluntary part-time employment. We add to the analysis by highlighting the role of gig employment in accelerating the flows out of involuntary part-time employment. Our paper can be seen as highlighting two important mechanisms: i) the changes in local labor demand conditions following the introduction of platform-based gig opportunities such as UberX, and ii) the changes in labor supply decisions influencing the worker flows from involuntary part-time employment.[3]

The rest of the paper is structured as follows. Section 2 provides a description of the data sources, sample creation, and summary statistics. In Section 3, we present the empirical strategy used in our analysis. Section 4 presents and discusses the results of our main analysis along with explorations of heterogeneity by treatment timing. In Section 5, we present the results of exercises highlighting the robustness of our findings. Section 6 presents a discussion of our results and puts them in the context of related work. Section 7 concludes the paper.

2. Data

2.1. Sources and Sample Creation

To investigate how the rollout of UberX affected involuntary part-time employment, we combine data from the Current Population Survey (CPS) with information on the dates of UberX entry across metareas. The CPS is a monthly survey of about 60,000 nationally representative U.S. households that collects information on employment status, labor force participation,

[3]Our results also are consistent with the conclusion in Valletta and Van Der List [35] that structural changes, especially in the industry composition, played a significant role in the increased involuntary part-time employment in the post-recession years.

unemployment, and demographic characteristics. The CPS records respondents' biological sex using binary male/female categories. We use this information to construct demographic controls at the metarea level, calculating the share of the labor force that is female. Throughout this paper, we use "female" and "male" to refer to these biological sex categories as recorded in the CPS. We acknowledge that these binary categories do not capture gender identity or the experiences of gender-diverse individuals. Importantly, the CPS is the only dataset that provides consistent information on involuntary part-time employment. We use the CPS basic monthly series from 2010 to 2019, extracted from IPUMS (Flood et al. [36]). Data on the UberX rollout comes from Hall et al. [11], who compiled city-level entry dates using official Uber blog announcements. This allows us to exploit both temporal and geographic variation in UberX expansion for causal analysis.

A limitation of CPS data, highlighted in prior research (Collins et al. [3]; Abraham et al. [14]; Abraham et al. [37] ; Garin et al. [9]; Jackson [10]), is that it primarily records respondents' main occupation. Because gig work often functions as a secondary job, the CPS cannot directly identify most gig workers. However, CPS remains the only survey with detailed measures of involuntary part-time employment, making it critical for our question. A second limitation is that CPS monthly estimates can be noisy at small geographic levels, such as counties or smaller metareas.

To address these issues, we construct our sample in three steps. First, we aggregate individual-level CPS data to the metarea-year level using survey weights. This bypasses the need to measure gig participation directly and instead captures changes in labor force outcomes at the metarea level, which we argue is the appropriate unit of exposure to UberX entry. Larger geographic aggregation also reduces sampling variability. Second, we aggregate monthly estimates to annual measures. Pooling across months improves precision and allows us to focus on longer-run dynamics, since the adjustment to UberX entry likely occurs gradually at the metarea level. Third, we restrict our sample to metareas with stable geographic boundaries and available data across the 2010–2019 period. This results in a balanced panel of 2,070 metarea-year observations.

Throughout our analysis, we employ unweighted population regressions, assigning equal weight to each metropolitan area. This approach reflects our research question about how the average metarea labor market responds to the expansion of the gig economy. Since UberX rollout decisions were made at the city-wide/metropolitan level, this weighting scheme treats each unit of treatment assignment equally. Given potential heterogeneity in treatment effects by metro size, our unweighted approach identifies the average effect across metros rather than conflating this with compositional differences across metro sizes.

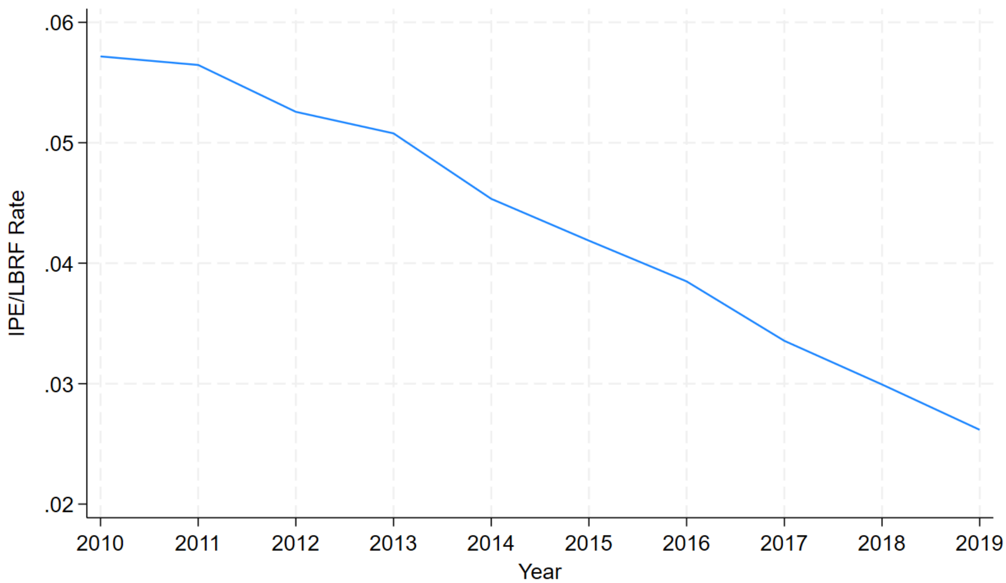
2.2. Involuntary Part-Time Trends and UberX Rollout

Similar to the unemployment rate, the involuntary part-time rate tends to be counter-cyclical, as employers reduce the number of hours they provide to employees during economic downturns. Due to the limited nature of outside job prospects during recessions, employees often want to work more hours but are unable to find employment that suits their desired level of labor. Being involuntary part-time employed is strongly correlated with the likelihood of higher unemployment durations, higher chances of being below the federal poverty line, and lower wage growth as compared to full-time workers [28, 29].

To provide an overview of recent trends in involuntary part-time employment, we present two variables: (i) involuntary part-time employment as a fraction of the total labor force, and (ii) involuntary part-time employment as a share of overall part-time employment throughout our sample. As presented in Figure 1, the share of the total labor force that is involuntary part-time employed was high during the beginning of our sample time period as a consequence of the Great Recession and, with the passage of time, experienced a gradual decline, reaching around 2.8 percent by 2019. In contrast, we find that part-time employment as a fraction of the total labor force remained fairly stable during the time period of our analysis, fluctuating between 22 percent and 24 percent (see Figure A.1 in Appendix A.3). This suggests that there was a change in the composition of part-time workers towards voluntary part-time employment during this time period, and not to other labor market statuses such as full-time employment or unemployed. Figure 2 illustrates this mechanism, showing a sharp decline in the share of part-time employment that is involuntary from approximately 25 percent in 2010 to 12 percent in 2019.

Over the same period of the 2010s, there was the introduction and a rise in popularity of platform-based "gig" work. Characterized by low entry costs and high flexibility in terms of hours and location, gig work primarily serves as a supplement to other stable, permanent jobs (Hall and Kruger [4]). Participation in this new type of gig employment, particularly in platform-based transportation gig work, increased rapidly throughout the mid-2010s.

Figure 1: Involuntary Part-time Employment as a fraction of Total Labor Force

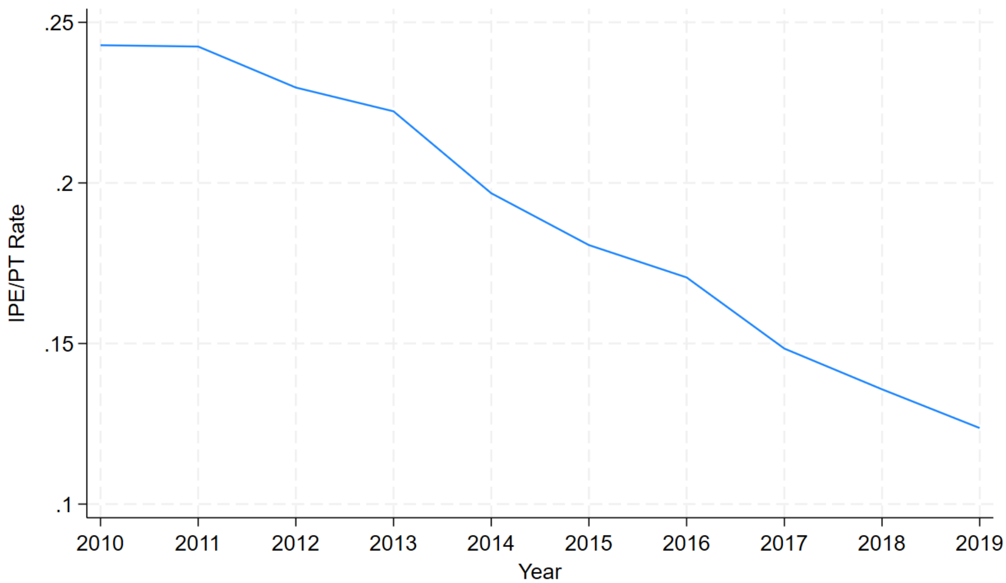


Note: This figure presents the share of the labor force in involuntary part-time employment. The involuntarily part-time employed are part-time workers who typically work fewer than 35 hours per week but would prefer a full-time position and are unable to obtain one due to economic conditions. Data come from the Current Population Survey individual monthly files aggregated to the metropolitan area-year level. The figure displays the unweighted average across the 207 metropolitan areas, assigning each metro an equal weight. Individual-level CPS data are aggregated to the metropolitan area-year level using survey weights to ensure each metro-year observation is representative of that area's population. The sample is restricted to metropolitan areas with stable geographic boundaries and data availability from 2010 to 2019.

Garin et al. [9] finds that the share of the labor force engaging in platform-based transportation gig work increased by more than 1 percentage point from 2013 to 2017, which corresponds to approximately 2 million workers. Our primary objective is to investigate the critical question of how the speed of recovery of involuntary part-time employment varies across metareas in response to exposure to UberX.

For our analysis, we consider UberX for several reasons. UberX employ-

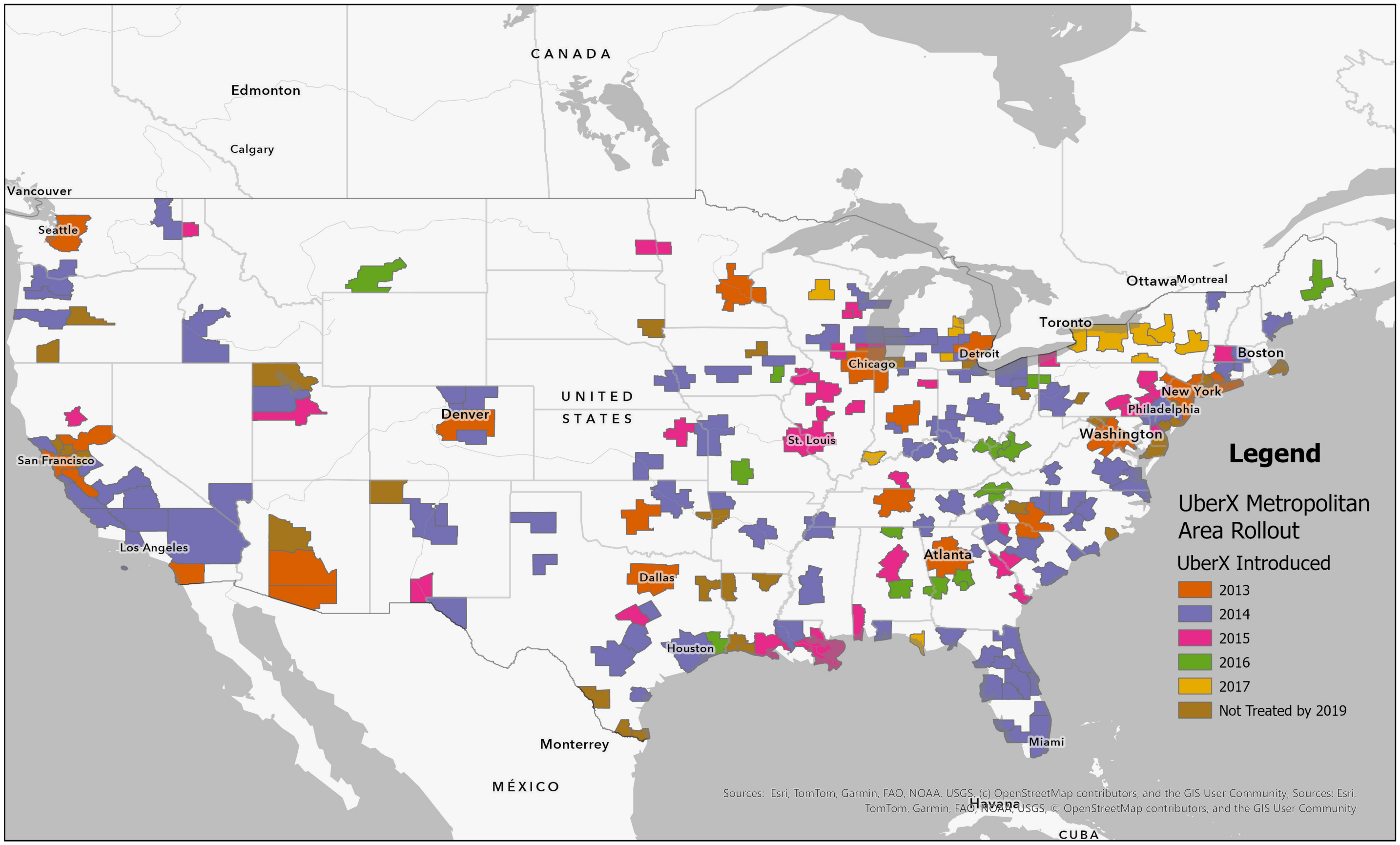
Figure 2: Involuntary Part-time Employment as a fraction of Total Part-Time Employment



Note: This figure presents involuntary part-time employment as a share of total part-time employment. The involuntarily part-time employed are part-time workers who typically work fewer than 35 hours per week but would prefer a full-time position and are unable to obtain one due to economic conditions. Data come from the Current Population Survey individual monthly files aggregated to the metropolitan area-year level. The figure displays the unweighted average across the 207 metropolitan areas, assigning each metro an equal weight. Individual-level CPS data are aggregated to the metropolitan area-year level using survey weights to ensure each metro-year observation is representative of that area's population. The sample is restricted to metropolitan areas with stable geographic boundaries and data availability from 2010 to 2019.

ment, as an Uber driver, serves as a well-known example of gig employment and is very popular among gig workers. UberX is Uber's standard ride-sharing service, where driver-partners benefit from a low entry barrier to employment and enjoy flexibility in both work hours and the location of their operations. Second, as highlighted in Shontell [38] and Scott [39], Uber is one of the first gig ride-hailing companies (founded in 2009) and has been a critical contributor to the rise in gig employment throughout the 2010s.

Figure 3: Rollout of UberX across Metareas by Year



Note: The map above represents the rollout of UberX across the 207 metropolitan areas in the sample, based on data from Uber blog announcements collected by Hall et al. [11]. Rollout dates were determined by the announcement date of the first service in each metropolitan area. Colors correspond to the year that UberX service initially entered the area. Metropolitan areas labeled 'Not Treated by 2019' had not received UberX service by the end of our sample period. Map lines delineate study areas and do not necessarily depict accepted national boundaries.

Table 1: Metarea Exposure to UberX rollout by year

First Treatment Year	Number of Metareas	Percent of Sample	Cumulative Percent
2013	19	9.18	9.18
2014	95	45.79	55.07
2015	35	16.91	71.98
2016	15	7.25	79.23
2017	11	5.25	84.54
Never Treated	32	15.46	100.00
Total	207	100.00	100.00

Notes: This table presents the distribution of UberX rollout timing across the 207 metropolitan areas in the sample. Rollout dates are based on data from Uber blog announcements collected by Hall et al. [11]. 'First Treatment' indicates the year UberX service was first introduced in each metropolitan area. 'Never Treated' represents the 32 metropolitan areas that had not received UberX service by the end of the sample period in 2019. The sample is restricted to metropolitan areas with stable geographic boundaries and data availability from 2010 to 2019.

It also represents a significant market share in the gig ride-hailing industry throughout our sample period. As mentioned in Kaczmarski [40], Uber had a market share of between 68 and 74 percent in the ride-hailing industry at the end of our sample period in 2017.

The rollout of UberX to various metareas throughout the 2010s is illustrated in Figure 3. UberX was introduced into major cities, including San Francisco, New York, Atlanta, and Chicago, in 2013. Following the initial deployment in 2013, the rollout pattern appears to target other large cities, including Houston, Tampa, Cleveland, and Milwaukee, in 2014. After 2014, the pattern of rollout is unclear, as metareas such as Green Bay and Bangor, as well as areas outside of Atlanta, are treated in the mid-to-late 2010s. The rate at which these metareas received UberX is shown in Table 1. In 2013, a small fraction of metareas were exposed to UberX, with approximately 19 of the 207 metareas, or around 9.0 percent of metareas, becoming exposed. However, in 2014, there was a significant rise, with 45.8 percent of the metareas being introduced to UberX. In the subsequent years, the rate of exposure decreased.

2.3. Summary Statistics

Table 2 reports descriptive statistics for metareas by UberX adoption group in 2012, the year preceding the initial rollout, and in 2019, the last year of the sample period. We characterize metareas into four groups: early adopters (metareas experiencing the rollout of UberX in 2013), middle adopters (metareas experiencing the rollout of UberX between 2014 and 2015), late adopters (metareas experiencing the rollout of UberX between 2016 and 2017), and never adopters (metareas that did not experience the rollout of UberX by 2019).

In 2012, labor force characteristics were broadly similar across groups, consistent with comparable pre-treatment trends, which are necessary for identification. Approximately 5 percent of the labor force was involuntary part-time employed, while 22–25 percent of part-time workers were involuntary. Part-time employment accounted for 22–24 percent of the labor force, and unemployment was roughly 8–9 percent.

By 2019, involuntary part-time employment had declined across all groups. The IPE share fell to about 2 percent in early and middle adopters, compared to 3 percent in late and never adopters. Among part-time workers, the involuntary share declined to around 12 percent in adopting groups, compared to 15 percent among never adopters, suggesting that UberX entry contributed to sharper reductions in adopting areas.

Observing the population demographic composition across the metareas, early adopters tended to have a slightly younger and more educated population compared to later adopters and non-adopters, while the gender composition was nearly identical across all adoption groups. Late adopters exhibited a slightly higher share of black residents. These demographic differences persisted through 2019, though the overall trends were stable.

All adoption groups exhibited similar labor market characteristics prior to UberX's entry, but diverged thereafter, with adopting areas, particularly early adopters, experiencing sharper reductions in involuntary part-time employment. In the next section, we present our empirical strategy for exploring the causal relationship between UberX adoption and involuntary part-time employment across metareas in the United States.

Table 2: Metarea Raw Means by Adoption Group: 2012 vs 2019

	Early		**Middle**		**Late**		**Never**	
Variable	(2012)	(2019)	(2012)	(2019)	(2012)	(2019)	(2012)	(2019)
Panel A: Labor Force Characteristics								
Labor Force Composition								
Full-Time Share	0.68	0.73	0.66	0.72	0.64	0.70	0.64	0.69
Part-Time Share	0.22	0.20	0.23	0.21	0.24	0.23	0.24	0.22
Unemployment Share	0.08	0.03	0.08	0.04	0.24	0.04	0.09	0.05
Involuntary Part-Time Employment								
IPE Share	0.05	0.02	0.05	0.02	0.05	0.03	0.06	0.03
IPE / Part-Time Share	0.25	0.12	0.22	0.12	0.22	0.12	0.25	0.15
Panel B: Metarea Population Demographics								
Age Composition								
Under 18	0.24	0.22	0.23	0.22	0.23	0.21	0.25	0.22
Age 18–24	0.09	0.09	0.10	0.09	0.09	0.09	0.10	0.08
Age 25–34	0.14	0.15	0.13	0.13	0.12	0.14	0.12	0.13
Age 35–44	0.14	0.14	0.13	0.12	0.12	0.11	0.12	0.11
Age 45–54	0.14	0.13	0.14	0.12	0.14	0.12	0.14	0.12
Age 55–64	0.12	0.12	0.13	0.13	0.14	0.14	0.12	0.13
Over 65	0.12	0.14	0.14	0.17	0.15	0.18	0.14	0.19
Demographic Composition								
Female	0.51	0.51	0.51	0.51	0.52	0.51	0.51	0.52
Black	0.13	0.13	0.12	0.12	0.14	0.14	0.11	0.09
Education Composition								
Below High School	0.13	0.11	0.12	0.11	0.14	0.10	0.14	0.13
High School	0.19	0.18	0.23	0.22	0.27	0.26	0.24	0.26
Some College	0.15	0.14	0.16	0.15	0.14	0.15	0.16	0.15
College	0.33	0.38	0.29	0.33	0.26	0.32	0.25	0.28
Group Size								
Number of Metareas	19		130		26		32	

Notes: Early adopters were first treated in 2013; Middle adopters in 2014-2015; Late adopters in 2016-2017; Never adopters were not treated by 2019. The table presents the raw means for 2012 (the first year) and 2019 (the last year) of the sample period. Data come from the Current Population Survey aggregated to the metropolitan area-year level. The table displays unweighted averages across metropolitan areas within each adoption group, assigning equal weight to each metarea. Rates are shares of the metro area labor force unless otherwise indicated. IPE share is the fraction of the labor force employed involuntarily as part-time workers. IPE/PT is the share of part-time employment that is involuntary in nature. Group sizes are constant across years.

3. Empirical Strategy

To estimate the impact of the UberX rollout on involuntary part-time employment, we use the staggered difference-in-differences estimator by Callaway and Sant'Anna [12].

Specifically, we compute group-time average treatment effects on the treated (ATTs) as

$$\mathrm{ATT}(g,t) = E\big[Y_{m,t}(1) - Y_{m,t}(0) \,|\, G_m = g,\, t \geq g,\, X_{m,t}\big]. \tag{1}$$

In the above specification, $\mathrm{ATT}(g,t)$ measures the average treatment effect of UberX entry in year t for metareas m that first adopt UberX in year g.

The outcome variable $Y_{m,t}$ represents one of our dependent variables of interest at the metarea-year level, including the involuntary part-time share of the labor force and involuntary part-time employment as a share of total part-time employment. As a robustness exercise, we also run this specification with the voluntary part-time share of the labor force, voluntary part-time employment as a share of total part-time employment, and other variables, including full-time employment as a share of the labor force, part-time employment as a share of the labor force, and the unemployment rate.

This approach generalizes the traditional two-way fixed effects (TWFE) estimator but avoids its problematic weighting. As shown by Goodman-Bacon [41], TWFE estimates are biased when treatment effects vary across groups or over time. This is particularly relevant in our setting: UberX adoption occurred across a five-year period during which the platform experienced dramatic growth in users, trips, and infrastructure. Metropolitan areas treated in 2013 faced a nascent platform, while those treated in 2016 encountered an established, rapidly expanding service, suggesting substantial heterogeneity in treatment intensity across adoption cohorts. The Callaway and Sant'Anna [12] estimator instead identifies group-time ATTs, which can be aggregated flexibly to recover static average treatment effects, dynamic event-study effects, and heterogeneity by treatment cohort.

Our causal identification relies on four assumptions: (i) absent UberX entry, the average change in labor force outcomes would have been the same for treated and control metareas; (ii) no spillovers of UberX entry across metareas; (iii) no anticipatory behavior prior to entry; and (iv) UberX rollout timing is uncorrelated with contemporaneous shocks to labor force outcomes.

Under these assumptions, the estimated ATTs capture the causal impact of UberX entry.

We aggregate the group-time ATTs in several ways. First, we estimate event-study specifications to test for pre-treatment parallel trends and to examine dynamic treatment effects. Second, we compute static ATTs, analogous to a conventional 2X2 DiD estimate, which serve as our main results. Finally, we explore heterogeneity across treatment cohorts by aggregating ATTs separately by adoption year.

In our specification, the comparison group consists of both never-treated and not-yet-treated metareas. Table 1 summarizes the composition of comparison groups by treatment cohort. As a robustness exercise, we also consider specifications that restrict the control group to never-treated areas only, and we vary the set of demographic and educational controls. The results reported in Appendix B are broadly consistent with our main estimates, although there is some variation in statistical precision. In addition, we conduct descriptive individual-level analyses (presented in Appendix A) to explore potential individual mechanisms of UberX exposure within these metareas, including changes in the probability of being involuntary/voluntary part-time employed, working hours, and household income.

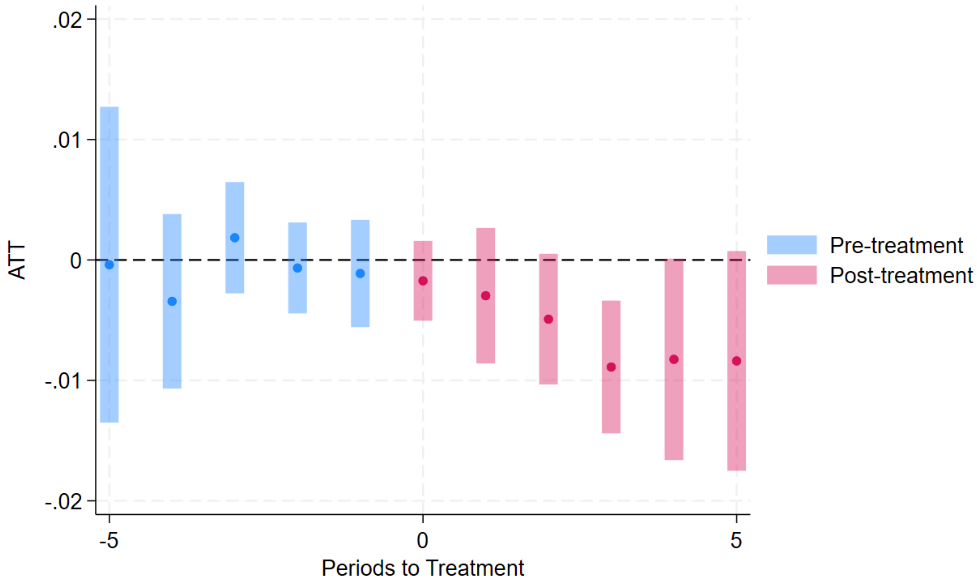
4. Results

4.1. Metarea Event Studies

Figures 4 and 5 present the event study estimates of UberX entry on involuntary part-time employment. The outcomes are (i) the share of the labor force that is involuntary part-time, and (ii) the share of part-time employment that is involuntary. These figures report results from the specification with demographic and educational controls, using not-yet-treated metareas as the comparison group. The results are consistent across comparison group definitions. The estimates are smaller in magnitude and less precise without demographic and age controls, but they become more stable and statistically significant when these controls are included. This motivates the use of the preferred specification shown in Figures 4 and 5.

The staggered difference-in-differences design relies on the identifying assumption that, in the absence of UberX entry, treated and not-yet-treated metareas would have followed parallel trends in involuntary part-time employment. The event study results provide evidence consistent with this assumption. We utilize a varying base period, as provided in the Callaway

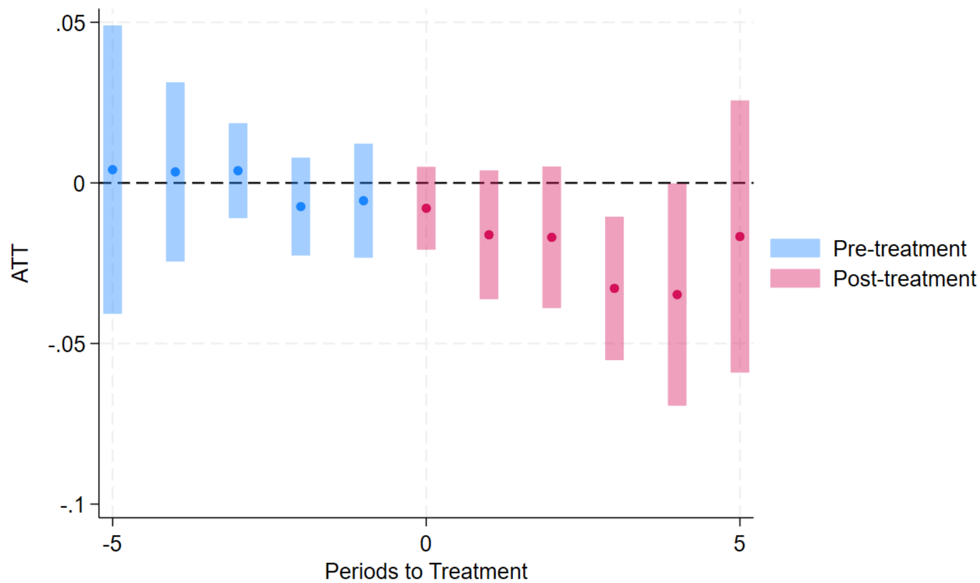
Figure 4: Event Study – Involuntary Part-Time Employment as a Share of Labor Force



Notes: This figure presents event study estimates of the effect of UberX rollout on involuntary part-time employment as a share of the labor force. Each point represents a treatment effect coefficient relative to UberX entry (period 0), with 95% confidence intervals. Estimates are obtained using the [12] estimator, which uses a varying base period, comparing outcomes in each period to those in the immediately preceding period. This approach produces estimates with confidence intervals for all periods, including pre-treatment periods, allowing for testing of parallel trends. Not-yet-treated metropolitan areas serve as the control group. Data come from the Current Population Survey aggregated to the metropolitan area-year level for 207 metropolitan areas from 2010-2019. Regressions are unweighted, giving each metropolitan area equal weight. Regressions include demographic, age, and education controls. Standard errors are clustered at the metropolitan area level.

and Sant'Anna [12] estimator package, which allows us to compare outcomes in each period to the immediately preceding period, thereby enabling the estimation of confidence intervals for all pre-treatment periods. In both outcomes, the pre-treatment coefficients are small in magnitude, statistically indistinguishable from zero, and show no systematic upward or downward trend.

Figure 5: Event Study – Involuntary Part-Time Employment as a Share of Part-Time Employment



Note: This figure presents event study estimates of the effect of UberX rollout on involuntary part-time employment as a share of the part-time employment. Each point represents a treatment effect coefficient relative to UberX entry (period 0), with 95% confidence intervals. Estimates are obtained using the [12] estimator, which employs a varying base period, comparing outcomes in each period to those in the immediately preceding period. This approach produces estimates with confidence intervals for all periods, including pre-treatment periods, allowing for testing of parallel trends. Not-yet-treated metropolitan areas serve as the control group. Data come from the Current Population Survey, aggregated to the metropolitan area-year level, for 207 metropolitan areas from 2010 to 2019. Regressions are unweighted, giving each metropolitan area equal weight. Regressions include demographic, age, and education controls. Standard errors are clustered at the metropolitan area level.

Moreover, there is no evidence of anticipatory effects immediately prior to UberX entry. Together, these patterns support the parallel trends assumption and strengthen the credibility of the empirical strategy.

Turning to the post-treatment dynamics, both outcomes display persistent declines following UberX entry. The decline in involuntary part-time employment as a share of the labor force (Figure 4) is more drastic one to two years after entry and remains statistically significant from three to four years post-treatment. A similar pattern is observed for involuntary part-time employment as a share of total part-time employment (Figure 5), with statistically significant declines beginning in year three and persisting through year four. In addition, Appendix A presents similar event studies on voluntary part-time employment, both as a percentage of the labor force and as a percentage of part-time employment. These results mirror the IPE results. These dynamic effects will be discussed further in the context of the static ATT in the next section and in the context of UberX expansion in the discussion section.

4.2. Metarea Results

Table 3 presents the static average treatment effects estimated from equation (1). In Panel A, we report the effect of UberX entry on the share of the labor force employed involuntarily part-time. In the baseline specification without controls, the coefficient is -0.003, corresponding to a 0.3 percentage point decline relative to non-exposed metareas. This estimate is not statistically significant. Adding age and demographic controls increases the magnitude to -0.004, which is significant at the 10 percent level. In the specification, which includes age, demographic, and educational controls, the coefficient is -0.005 and is statistically significant at the 5 percent level. Relative to the 2012 pre-treatment mean of 0.05 (5 percent), this estimate implies a 10 percent decline in the share of the labor force employed involuntarily part-time in metareas exposed to UberX.

Panel B of Table 3 reports the effects on the share of part-time employment that is involuntary. The baseline specification indicates a decline of 1.7 percentage points, statistically significant at the 5 percent level. The inclusion of age and demographic controls yields a similar magnitude and significance. In the specification with age, demographic, and educational controls, the effect is -0.020, corresponding to a 2.0 percentage point reduction, which is statistically significant at the 1 percent level. Relative to the 2012 pre-treatment mean of 0.23 (23 percent), this corresponds to an 8.7 percent decline in the share of part-time workers employed involuntarily.

These results are generally robust to the use of an alternative comparison group. Appendix B reports estimates using the never-treated metareas as the control group. The magnitudes are nearly identical to those shown in Table 3. The only exception is that, when controlling for age, education, and

Table 3: Static Average Treatment Effects on Involuntary Part-Time Employment

Panel A: Involuntary Part-Time as a Share of the Labor Force

	(1) No Controls	(2) Age	(3) Age + Demo	(4) Full
ATT	-0.003	-0.003	-0.004*	-0.005**
	(0.003)	(0.002)	(0.002)	(0.002)
Observations	2,070	2,070	2,070	2,070
2012 Mean	0.05	0.05	0.05	0.05

Panel B: Involuntary Part-Time as a Share of Part-Time Employment

	(1) No Controls	(2) Age	(3) Age + Demo	(4) Full
ATT	-0.017**	-0.014*	-0.017**	-0.020***
	(0.008)	(0.008)	(0.008)	(0.008)
Observations	2,070	2,070	2,070	2,070
2012 Mean	0.23	0.23	0.23	0.23

Notes: This table presents static average treatment effects (ATT) of UberX entry on involuntary part-time employment under different control specifications. Panel A uses IPE as a share of the labor force as the dependent variable; Panel B uses IPE as a share of part-time employment. Estimates are obtained using the [12] estimator. Column (1) includes no controls; (2) includes age composition controls; (3) adds demographic controls (sex, race); (4) adds education controls. Data come from the Current Population Survey, aggregated to the metropolitan area-year level, for 207 metropolitan areas from 2010 to 2019. Regressions are unweighted, giving each metropolitan area equal weight. The 2012 Mean shows the pre-treatment baseline mean of the dependent variable. Not-yet-treated metropolitan areas serve as the control group. Standard errors, clustered at the metropolitan area level, are shown in parentheses. Observations represent metropolitan area-year combinations. *, **, *** denote significance at the 10%, 5%, and 1% levels, respectively.

demographics, the estimate loses statistical significance, although it remains similar in magnitude.

4.3. Heterogeneity Analysis

In this subsection, we investigate the impact of UberX on involuntary part-time employment by examining the timing of treatment. Cohort-specific event studies are provided in Appendix A and show that pre-treatment coefficients are generally close to zero and not statistically significant, with a few exceptions. Table 4 reports static ATT estimates by treatment cohort. A consistent pattern is evident for both outcomes, as the effects are largely concentrated among early-adopting metareas.

For involuntary part-time employment as a share of the labor force, the 2013 and 2014 cohorts experienced declines of 0.7 and 0.8 percentage points, respectively. The 2014 effect is statistically significant at the 1 percent level, while the 2013 effect is not statistically significant at conventional levels. Later cohorts show weaker effects: the 2015 and 2017 cohorts display essentially null results, while the 2016 cohort shows a decline of 0.8 percentage points, similar in magnitude to the early cohorts but not statistically significant.

Turning to the share of part-time employment that is involuntary, the 2013 and 2016 cohorts show the largest effects, with declines of 4.1 and 4.2 percentage points, respectively; the 2013 result is statistically significant at the 5 percent level, while the 2016 result is not significant. The 2014 cohort also shows a statistically significant reduction of 2.3 percentage points. In contrast, the 2015 cohort shows no measurable effect, and the 2017 cohort has a small positive coefficient (1.7 percentage points), neither of which is statistically significant.

Figures A.4 and A.5 provide the dynamic treatment ATT's broken down by cohort adoption group. As can be seen from these figures, the effects on IPE for early adopters (2013-2014) were delayed to 2-to-3 years post-exposure of UberX, while the 2016 treatment group saw immediate IPE labor market effects. The most prominent effects were observed in the 2014 treatment group, which corresponds to the largest period of treatment in the sample (45.8

Overall, the treatment cohorts that are driving our main results are the early adopters (2013-2014), and the 2016 adopters. This pattern of results follows the rapid expansion of UberX from 2013 to 2019. As described in Appendix A, 2014 was the single largest year of UberX expansion, targeting many medium-sized metareas after the service had already entered the largest U.S. cities such as Chicago, New York, and San Francisco in 2013. However, the rapid expansion of demand for UberX services did not occur

Table 4: Static Average Treatment Effects by Treatment Cohort

Panel A: Involuntary Part-Time as a Share of the Labor Force

	Group	Treatment Cohort				
	Avg.	2013	2014	2015	2016	2017
ATT	-0.005**	-0.007	-0.008***	0.002	-0.008	0.001
	(0.002)	(0.004)	(0.003)	(0.003)	(0.011)	(0.007)
Observations	2,070	2,070	2,070	2,070	2,070	2,070
2012 Mean	0.05	0.05	0.05	0.05	0.05	0.05

Panel B: Involuntary Part-Time as a Share of Part-Time Employment

	Group	Treatment Cohort				
	Avg.	2013	2014	2015	2016	2017
ATT	-0.019**	-0.041**	-0.023**	-0.002	-0.042	0.017
	(0.008)	(0.016)	(0.011)	(0.014)	(0.040)	(0.026)
Observations	2,070	2,070	2,070	2,070	2,070	2,070
2012 Mean	0.23	0.23	0.23	0.23	0.23	0.23

Notes: This table presents static average treatment effects (ATT) of UberX entry by treatment cohort. Panel A uses IPE as a share of the labor force as the dependent variable; Panel B uses IPE as a share of part-time employment. The "Group Avg." report displays the overall ATT across all cohorts. Treatment cohorts indicate the year UberX first entered the metropolitan area. Estimates are obtained using the [12] estimator, which incorporates full demographic, age, and education controls. Data come from the Current Population Survey, aggregated to the metropolitan area-year level, for 207 metropolitan areas from 2010 to 2019. Regressions are unweighted, giving each metropolitan area equal weight. The 2012 Mean shows the pre-treatment baseline mean of the dependent variable. Not-yet-treated metropolitan areas serve as the control group. Standard errors, clustered at the metropolitan area level, are shown in parentheses. Observations represent metropolitan area-year combinations. *, **, *** denote significance at the 10%, 5%, and 1% levels, respectively.

until 2016, which would follow the pattern of delayed labor market effects for early adopters and immediate impacts for the 2016 treatment group. [4]

[4] According to internal Uber company data reported by Iqbal [13], the amount of Uber

While our main analysis focuses on metropolitan area-level outcomes, Appendix A explores descriptive individual-level patterns using the monthly CPS repeated cross-sections. Although we cannot establish causal individual-level mechanisms due to the data limitations discussed earlier, these descriptive results provide suggestive evidence consistent with our proposed mechanism. We find that individuals in metropolitan areas who have been exposed to UberX exhibit lower probabilities of involuntary part-time employment, higher probabilities of voluntary part-time employment, higher household incomes among voluntary part-time workers, and increased working hours among part-time workers. These patterns are consistent with workers transitioning from involuntary to voluntary part-time employment as UberX provides flexible earning opportunities. Full details on data construction and methodology are provided in Appendix A

5. Robustness

As discussed in the literature, most Uber drivers are part-time workers who use gig work as a supplement to other forms of permanent employment (Hall and Krueger [4], Garin et al. [9]). Building on this background, we test our main specification against a series of other labor market outcomes: the share of the labor force in voluntary part-time employment, the share of part-time workers in voluntary part-time employment, the share of the labor force working full-time, the share of the labor force working part-time, and the unemployment rate. This exercise serves two purposes. First, the full-time share, part-time share, and unemployment rate serve as placebo outcomes: if UberX primarily affects involuntary part-time workers, we should not see systematic changes in these other labor force components. Second, examining voluntary part-time outcomes allows us to explore whether UberX may have facilitated transitions from involuntary to voluntary part-time work.

capital funding received by the company grew from 0.44 billion in 2013 to 4.64 billion in 2014 to 9.14 billion in 2016. The number of Uber users increased from 11 million in 2015 to 37 million in 2016 and further to 58 million in 2017. During the same period, revenue from gross bookings surged from $ 19 billion in 2016 to $ 45 billion in 2017. Finally, the number of Uber trips increased from 3.79 billion in 2017 to 5.21 billion in 2018.

Table 5: Static Average Treatment Effects on Placebo Outcomes

Panel A: Voluntary Part-Time Outcomes

	VPE Labor Force Share	VPE Part-Time Share
ATT	-0.000	0.020***
	(0.003)	(0.008)
Observations	2,070	2,070
2012 Means	0.179	0.770

Panel B: Other Labor Force Composition Outcomes

	Full-Time Share	Part-Time Share	Unemployment Rate
ATT	0.001	-0.006	0.000
	(0.005)	(0.004)	(0.002)
Observations	2,070	2,070	2,070
2012 Means	0.655	0.232	0.081

Notes: This table presents static average treatment effects (ATT) of UberX entry on placebo outcomes, estimated using the specification with full demographic, age, and education controls. Panel A examines the outcomes of voluntary part-time employment, where voluntary part-time employees are those who work part-time by choice. Panel B examines other labor force composition outcomes. Estimates are obtained using the Callaway and Sant'Anna (2021) estimator. Data come from the Current Population Survey, aggregated to the metropolitan area-year level, for 207 metropolitan areas from 2010 to 2019. Regressions are unweighted, giving each metropolitan area equal weight. The 2012 Means show the pre-treatment baseline mean of each dependent variable. Not-yet-treated metropolitan areas serve as the control group. Standard errors, clustered at the metropolitan area level, are shown in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% levels, respectively. Observations represent metropolitan area-year combinations.

Table 5 reports results from our specification, which includes age, demographic, and educational controls. Starting with Panel A, the voluntary part-time outcomes, we find mixed results. The share of part-time workers in voluntary part-time employment increases by 0.020 (2.0 percentage points), which is significant at the 1 percent level. This mirrors the 2.0 percentage point decline in involuntary part-time employment as a share of part-time work, as documented in Table 3, consistent with the interpretation that some involuntary part-time workers shifted into voluntary part-time status after UberX's entry. By contrast, the voluntary part-time share of the labor force shows a small and statistically insignificant effect. Event studies for the voluntary part-time outcomes are available in Appendix A.

Turning to Panel B, which reports results for the share of the labor force working full-time (full-time share), the share of the labor force working part-time (part-time share), and the unemployment rate, the estimates are small and statistically insignificant. The full-time share and the unemployment rate are essentially unchanged in metareas exposed to UberX relative to those that were not.

The part-time share shows a modest decline of -0.006 (0.6 percentage points), but this estimate is not statistically significant at conventional levels. These results are consistent with prior literature, which suggests that gig work is primarily undertaken by part-time workers rather than full-time workers or workers who are unemployed.

Overall, these placebo and complementary outcomes strengthen our main findings. We do not find evidence that UberX affected broader measures of labor force composition. However, the mirror pattern between involuntary and voluntary part-time employment supports the interpretation that the introduction of gig opportunities facilitated worker transitions within part-time employment.

We further assess the validity of our results using randomization inference. Specifically, we randomly permute the timing of UberX entry across metareas 1,000 times and re-estimate the static ATT, generating a placebo distribution of treatment effects under the null of no treatment effect. The results of this randomization inference are shown in Table 6. For both primary outcomes, involuntary part-time employment as a share of the labor force and the share of part-time workers that are involuntary, the observed ATTs lie in the lower tail of the permutation distribution. The randomization p-values are 0.032 and 0.039, respectively, both of which are significant at the 5 percent level. These results reinforce the robustness of our specification, suggesting that the observed effects are unlikely to be due to chance treatment assignment.

Table 6: Randomization Inference Robustness Tests

	Observed ATT	Conventional p-value	RI p-value
IPE Share of Labor Force	-0.0055	0.010	0.032
IPE Share of PT Workers	-0.0204	0.006	0.039

Notes: This table reports randomization inference (RI) p-values testing whether the observed treatment effects could arise by chance. RI p-values are based on 1,000 permutations of UberX entry timing across metropolitan areas, with treatment timing randomly reassigned within clusters (metropolitan areas) to preserve the clustering structure. Conventional p-values are based on clustered standard errors at the metropolitan area level. Observed ATT values are point estimates from the specification with full demographic, age, and education controls. RI p-values indicate the proportion of placebo ATTs that are as extreme or more extreme than the observed ATT. Estimates are obtained using the [12] estimator. Data come from the Current Population Survey, aggregated to the metropolitan area-year level, for 207 metropolitan areas from 2010 to 2019. Regressions are unweighted, giving each metropolitan area equal weight. Not-yet-treated metropolitan areas serve as the control group. IPE Share of Labor Force refers to the fraction of the total labor force working involuntary part-time. IPE Share of PT Workers refers to the share of part-time workers who are involuntary.

In addition to the placebo outcomes and randomization inference, we also conduct several other robustness checks. First, we re-estimate our models using an alternative comparison group of never-treated metareas. The results, presented in Appendix B, are of similar magnitude to those in our original specification, which includes age, education, and demographic controls. Together, these robustness exercises further strengthen the validity of our main findings.

6. Discussion

The main question from our estimates is whether they are reasonable magnitudes for the impact of UberX on metarea labor forces. Collins et al. [3] finds that, from the universe of tax returns, the share of the workforce with income from gig work increased by about 1.9 percentage points between 2000 and 2016, with more than half of the increase occurring between 2013 and 2016. Garin et al. [9] shows that roughly 12 percent of the 2018 workforce engaged in contract-based gig work, and the share participating in transportation-app gig work grew from nearly zero in 2013 to about 1 percent of the labor force (by over 2 million workers) by 2017. They also find that earnings from transportation apps are small and supplemental to other wages, concluding that the share of workers relying on gig work as a primary job has remained stable, with most engaging in it part-time.

According to Edwards and Smith [42], at the end of our sample, i.e., in 2019, the size of the labor force was 158.6 million people. Our results imply that exposure to UberX reduced the number of workers in the labor force who are involuntary part-time employed in a metarea by half a percentage point of the total labor force. With a back-of-the-envelope calculation, half a percentage point of the 2019 labor force results in a reduction of 794,300 involuntary part-time employed workers in the labor force. Combining this calculation with insights from the market share of UberX in ride-hailing (Kaczmarski [40]), the rise in transportation app gig work participation (Garin et al. [9]), the role of gig work as a supplemental job (Hall and Krueger [4]), and the predominance of part-time workers moving in and out of gig work (Garin et al. [9]), our findings are consistent with the literature if we assume that less than half of those entering transportation gig work were involuntary part-time workers who later self-identify as something other than involuntary part-time employed.

As observed in the previous section, we find statistically significant declines in both the involuntary part-time employed as a share of the total labor force and the involuntary part-time employed as a share of the total part-time employed following exposure to UberX. We also find that other labor force composition variables, such as the share of the labor force working full-time, the share of the labor force working part-time, and the unemployment rate, seem to remain unaffected by exposure to UberX. These results are consistent with the existing literature. Hall and Krueger [4] finds that only 8 percent of Uber drivers were previously unemployed before partnering with Uber, while 81 percent reported having permanent jobs alongside Uber. Garin et al. [9] conclude that the overall share of the workforce relying on transportation gig work as a primary job has remained relatively constant over time, suggesting that the flow of workers entering and exiting transportation gig work is those who are engaging with it on a part-time basis.

This suggests that gig work, particularly in transportation services like Uber, acts as a mechanism for shifting involuntary part-time workers into voluntary part-time roles rather than redistributing them elsewhere in the labor force. Due to the nature of observing involuntary part-time employment changes at a metarea level, we can only offer speculation on why gig work would achieve this. Gig work, compared to other employment opportunities, offers truly flexible labor supply decisions. Involuntary part-time workers self-indicate they want and are available for more hours of work, but either need to find additional part-time opportunities that can work with their current employer or find full-time work opportunities. Gig work offers a more accessible opportunity to earn labor hours, with low barriers to entry and a highly flexible schedule. This would shift a worker from involuntary to voluntary part-time employment, as gig work allows them to supplement their primary part-timejob.

To summarize, our analysis captures aggregate metarea-level responses in the labor market to the rollout of UberX services. However, a limitation of this paper is that it fails to evaluate how the introduction of UberX affects workers differently across key demographic characteristics such as age, sex, and race. This is primarily due to the limited scope of the CPS dataset in capturing such individual-level gig work participation.[5] In this regard, administrative data, such as IRS tax records, offer larger samples and more direct measures of gig earnings, making them better suited for investigating these differential outcomes. Future research using individual-level administrative data could examine how gig work opportunities, such as UberX, affect involuntary part-time employment across demographic groups and enhance our understanding of these heterogeneous effects.

7. Conclusion

This paper examines the impact of the rollout of platform-based gig opportunities on involuntary part-time employment in the United States. Focusing on the period from 2010 to 2019, we present causal evidence indicating that the gradual introduction of UberX services has led to a decline in involuntary part-time employment across metareas in the United States. Utilizing the quasi-random nature of the rollout of UberX services across metareas, we first provide estimates of a reduction of about 10 percent of involuntary part-time employment as a share of the labor force. Expressed as a fraction of total part-time employment, the involuntary component declines by 8.3

[5]Although the CPS provides binary sex indicators used in our metarea-level controls, analyzing sex-based heterogeneity in treatment effects requires individual-level data.

percent when UberX becomes available across metareas. This is in perspective with the fact that no other labor market variables, such as the share of the labor force working part-time, full-time employment as a fraction of the total labor force and the unemployment rate, experience any movement as a consequence of the rollout of UberX. Thus, our findings indicate a mechanism of worker reallocation within part-time employment, shifting from the involuntary to the voluntary component, which can be primarily attributed to the additional flexibility in terms of work hours and operational location offered by ride-sharing platforms such as UberX.

Glossary

Average Treatment Effects on the Treated (ATT): The ATT is a type of causal effect that estimates the effect of receiving a treatment (or intervention policy) on the subset of participants who actually received treatment in the study.

Event Study: A graphical analysis showing treatment effects over time relative to the treatment date. Typically used to examine dynamic treatment effects and visualize the parallel trends assumption.

Gig Work: Jobs in which independent contractors, online platform workers, contract firm workers, on-demand workers, and temporary workers enter formal agreements to provide on-demand services on behalf of a company. A popular type of gig work is transportation gig work, and includes companies such as Uber, Lyft, and Rideshare.

Group-Time ATT: As defined by Callaway and Sant'Anna [12] as the average treatment effect for group g at time t, where a "group" is defined by the time period when units are first treated.

Intention-to-Treat (ITT): The effect of treatment assignment for all participants who are analyzed according to the original group they were assigned, regardless of what treatment they actually receive.

Involuntary Part-Time Employment (IPE): Defined by the Bureau of Labor Statistics as a category of people who indicated that they would like to work full time but were working part time (1 to 34 hours) because of economic reasons, such as cut hours or being unable to find full-time jobs.

Labor Market Slack: Labor market slack is an economic situation where there are more job seekers than available jobs, creating excess quantity supplied of labor.

Metropolitan Area: Also known as metarea. This region comprises a large urban core, along with surrounding communities that exhibit a high degree of economic and social integration. Metropolitan areas are defined by the Office of Management and Budget.

Parallel Trends Assumption: Necessary condition for causal identification in a Difference-in-Differences analysis. The condition states that the trends in the outcome of interest for the treated group without treatment intervention would have been the same as the observed trend in the control group.

Part-Time Employment: Defined by the Bureau of Labor Statistics in the Current Population Survey as workers who typically work fewer than 35 hours per week. This is in contrast to full-time workers, who are defined as those who typically work more than 35 hours per week.

Staggered Difference-in-Differences (DiD): An econometric method for causal inference in which treatment occurs at different times across units, allowing for treatment effect heterogeneity.

Two-Way Fixed Effects (TWFE): A statistical method utilizing panel data that controls for both unobserved unit-specific effects and time trend effects simultaneously.

Voluntary Part-Time Employment: Defined by the Bureau of Labor Statistics as a category of people who indicated that they usually work part-time for noneconomic reasons such as childcare, family or personal obligations, school or training, retirement, or Social Security limits on earnings, and other reasons such as preference.

Author Contributions

Matt Eaton: Conceptualization, Methodology, Formal Analysis, Investigation, Data Curation, Writing - Original Draft, Writing - Review & Editing, Visualization.
Srijan Banerjee: Conceptualization, Validation, Writing - Original Draft, Writing - Review & Editing, Visualization.

Declaration of Interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Data Statement

The data used in this study are publicly available. The labor market data used in this study are publicly available from IPUMS-CPS (Integrated Public Use Microdata Series - Current Population Survey). The data can be accessed at `https://cps.ipums.org/` Flood et al. [36]. Information on the UberX rollout timing across areas was compiled from publicly available sources, including Uber's blog, and was compiled by Hall et al. [11].

Acknowledgements

We are grateful to Jonathan Leganza, Devon Gorry, David Drukker, Curtis Simon, Aspen Gorry, Cheng Chen, Paulo Lins, Joseph Pedtke, participants of the Applied Theory Reading Group, Labor Economics and Macroeconomics Workshops at Clemson University and participants at the 2024 Southern Economic Association Conference and the 2025 Midwest Economic Association Conference for useful comments and feedback.

Declaration of Generative AI

During the preparation of this work, the author(s) used ChatGPT and Claude AI in order to assist with coding structure, manuscript grammar, structure, and formatting. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

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Appendix A. Extended Context & Supplemental Figures

Appendix A.1. History of Uber and Rollout of UberX

In March 2009, Travis Kalanick and Garrett Camp founded Uber in San Francisco, California. Following a beta release in May 2010, Uber officially launched later that year with its flagship black-car service, Uber Black, which allowed users to hail luxury vehicles driven by commercially licensed drivers at prices roughly 1.5 times higher than a taxi (McAlone [43]). In July 2012, Uber introduced UberX, initially permitting non-luxury vehicles but still requiring commercial licenses. In April 2013, UberX expanded to allow everyday drivers with personal vehicles, subject to background checks as well as requirements for insurance, registration, and minimum vehicle standards (Lawler [44]).

The transition from a luxury service to a peer-to-peer ride-sharing model enabled UberX to rapidly expand across the United States. Uber first targeted large metareas where conventional taxi demand was unmet and legacy systems constrained supply, such as New York City, Chicago, Boston, and Seattle. These cities offered large pools of potential riders and drivers, but were also characterized by regulatory frameworks, such as medallion systems, that limited the number of cabs. Entering such markets gave UberX a clear opportunity to capture unmet demand (Alonso [45]). Haldun and Ellison [46] find that Uber tended to enter cities where existing regulation distorted pre-existing taxi markets, allowing Uber to profit by circumventing regulatory entry barriers.

Uber entered these markets by operating without prior regulatory approval, adopting a "launch first and ask forgiveness later" strategy (Arora [47]). In practice, this meant quickly building a loyal rider and driver base and, when challenged by authorities, arguing that existing taxi regulations did not apply to Uber's business model. At the same time, Uber leveraged its growing user base as political capital, mobilizing customers and drivers to resist attempts at legal restrictions (Solon [48]).

Competition also shaped UberX's rollout. Peer-to-peer ride-sharing platforms like Lyft and Sidecar debuted in San Francisco in the summer of 2012, offering cheaper alternatives to traditional taxis and undercutting Uber Black's price point. This competition prompted Uber to rapidly expand UberX to align with its rivals' models and protect its market position (Gannes [49]). All of these companies sought a first-mover advantage by scaling quickly into U.S. cities, creating strong incentives for Uber to enter new markets before or at the same time as its competitors (Stone [50]).

The speed of Uber's U.S. expansion was fueled by unprecedented amounts of venture capital. By 2015, Uber had raised over $6.6 billion in funding dedicated to aggressive city launches, rider subsidies, and driver recruitment programs (Hoffman [51]). In new markets, Uber frequently offered free or heavily discounted rides to quickly establish a customer base (Hoffman [51]). Simultaneously, it recruited drivers with bonuses, referral bounties, and even financing and leasing programs to help those without eligible vehicles join the platform (Solon [48]). These tactics enabled Uber to rapidly establish itself in new markets, even at substantial financial losses.

As a result, Uber's footprint grew from a handful of major U.S. cities in the early 2010s to more than 250 cities worldwide by the mid-2010s. Expansion peaked in 2014, when Uber raised over $2 billion and was reportedly entering nearly one new city every week (Cantale and Hutton [52]). In the United States, UberX became available in essentially every major and mid-sized metarea within three years of its 2012 launch.

This pattern of rapid expansion is clearly evident in our sample of metareas, as shown in Table 1 and Figure 3. The initial rollout of UberX in 2013 occurred first in major cities such as San Francisco, Seattle, Chicago, New York, and Washington, D.C., totaling around 19 different metareas. Expansion increased significantly during 2014 to the next largest metareas, such as Madison, Milwaukee, Houston, and Tampa, coinciding with a large increase in investment capital into Uber. By the end of 2014, over 95 metareas had been exposed to UberX. Finally, UberX expanded to smaller metareas, such as New Orleans and St. Louis, throughout the latter part of the 2010s.

In summary, UberX's rollout was guided by three central strategies: (1) targeting markets with unmet transportation demand, often shaped by restrictive taxi regulations; (2) moving rapidly to outpace competitors and establish first-mover advantage; and (3) leveraging vast amounts of venture capital to subsidize growth, recruit drivers, and build a large user base. Together, these strategies enabled UberX to achieve explosive expansion across the United States during the 2010s.

Appendix A.2. Individual Analysis

In addition to our metarea analysis, we investigate the impact of the rollout of UberX on involuntary part-time employment at an individual level. The goal of this analysis is to understand and provide suggestive evidence on the individual-level characteristics that shaped the decline in involuntary part-time employment in the United States between 2010 and 2019.

For our analysis, we combine CPS monthly files into a pooled cross-sectional dataset. We focus on prime-age workers for whom information on location, demographics, education, household income, hours worked, and occupation is available from 2010 to 2019. We further restrict the sample to metareas with consistent geographic boundaries over the period, mirroring our approach in the metarea-level analysis. This results in around 4,206,620 observations. In addition to the full labor force sample, we examine individual outcomes separately for part-time workers, involuntary part-time workers, and voluntary part-time workers.

We begin our analysis by examining the impact of exposure to UberX on individuals in the pooled monthly CPS sample. Table A.1 provides an overview of demographics, educational, and occupational information grouped by involuntary part-time workers, part-time workers, and the overall labor force.

Involuntary part-time workers make up around 4 percent of the labor force in our sample, and around 18 percent of part-time workers. Additionally, they exhibit younger age profiles, higher shares of male workers, and disproportionately greater representation of Black and Hispanic workers relative to both the overall part-time employed sample and the broader labor force. They are also less formally educated on average and have lower household income compared to part-time workers and the overall labor force.[6] The top occupations for involuntary part-time workers are office and administration, food preparation, cleaning and maintenance, construction, transportation, and sales, respectively.

[6]Involuntary part-time workers tend to have higher usual weekly hours compared to part-time workers.

Table A.1: Demographic, Educational, and Occupational Characteristics

Variable	**IPE Workers**	**Part-Time**	**Overall Labor Force**
IPE	1	0.184	0.039
Demographics			
Age	37.649	39.878	41.84
Female	0.479	0.606	0.484
White	0.720	0.718	0.730
Black	0.178	0.126	0.129
Asian	0.012	0.008	0.055
Race 2 More	0.020	0.023	0.016
Hispanic	0.265	0.212	0.178
Education			
Less than HS	0.164	0.130	0.093
HS Degree	0.549	0.480	0.399
College Degree	0.249	0.355	0.408
Income and Hours Worked			
Household Income	$56,312.7	$78,421.5	$86,304.7
Usual Hours Worked per Week	39.4	26.62	39.98
Top Occupations			
Sales	0.046	0.132	0.104
Food Preparation	0.139	0.140	0.052
Office and Administration	0.215	0.208	0.234
Construction	0.066	0.034	0.070
Cleaning and Maintenance	0.083	0.061	0.079
Transportation	0.060	0.035	0.057
Observations	160,059	916,904	4,206,827

Notes: This table presents summary statistics from the 2012-2019 Current Population Survey individual monthly files, pooled as repeated cross-sections. The sample is restricted to individuals in the 207 metropolitan areas included in the analysis. Statistics are weighted using CPS survey weights to be representative of each population group. IPE Workers are those reporting involuntary part-time employment; Part-Time includes all part-time workers; Overall Labor Force includes all employed individuals in the sample. The IPE row indicates the share of each group that is involuntarily part-time (by definition 1 for IPE Workers). Observations represent individual-month observations. Income values are in 2019 dollars.

For the individual analysis, we estimate the following equation,

$$Y_{i,m,t} = \alpha + \beta_1 T_{m,t} + \Gamma X_{i,t} + \delta M_m + \eta Y_t + \epsilon_{i,m,t}. \tag{A.1}$$

In the above equation, $Y_{i,m,t}$ is the outcome variable, namely, the probability of being involuntary part-time employed, voluntary part-time employed, household income, or usual hours worked during the week, indexed by person, i, in metarea, m, in year,t. The independent variable of interest is the intention-to-treat (ITT),T, where treatment to UberX is at a metarea level,m, in year t. We control for a vector of demographic characteristics of the individual,X, such as gender, race, and education, and control for metarea and year fixed effects with M and Y, respectively.

In our estimation, β_1 captures the average intention-to-treatment effect (ITT) of being in a metarea exposed to UberX compared to individuals living in metareas not exposed to UberX, controlling for differences in metareas and across time. The identification of this ITT effect relies on utilizing variation in treatment within metareas over time, relative to changes in treated areas, essentially conducting a difference-in-difference analysis with the repeated cross-sections of individuals from the CPS monthly files. This identification strategy relies on the parallel trends assumption between treated and untreated metareas, and no anticipation of UberX exposure to individual i metarea that would induce a change in their behavior. We demonstrate later in Section 4 that the parallel trends assumption appears to hold between metareas exposed to UberX and those not exposed to UberX, and there appears to be no anticipatory behavior in metareas prior to their exposure to UberX.

The results of the above specification are shown in Table **??**. Panel A shows the impact of being in a metarea exposed to UberX on the probability of being involuntary part-time employed and the probability of being voluntary part-time broken down by the full labor force sample and the part-time sample. We find the average intention-to-treat effect of living in a metarea exposed to UberX results in a 0.1% decrease in the probability of being involuntary part-time employed compared to individuals in metareas not exposed to UberX in the sample utilizing the entire labor force. This effect increases in magnitude to around a half a percentage point decrease in the probability of being involuntary part-time employed when using only the sample of part-time workers. When we consider the probability of being voluntary part-time employed, we find results that closely mirror those for involuntary part-time employment. Considering the total labor force, the average intention-to-treat effect increases the probability of being voluntary part-time employed by 0.1%, although this increase is statistically insignificant. Focusing on part-time employed workers, living in a metarea exposed to UberX, increases the probability of being voluntarily part-time employed by around half a percentage point compared to part-time workers not living in a metarea exposed to UberX, and this difference is statistically significant at the 5% level. This complements our metarea level results, indicating a worker reallocation from involuntary to voluntary part-time employment, in response to exposure to UberX.

Table A.2: Intention-to-treat Exposure to UberX

Panel A: Probability of IPE/VPE				
Outcome:	Prob(IPE)		Prob(VPE)	
Coefficient	-0.001**	-0.004**	0.001	0.005**
(Std. Error)	(0.000)	(0.002)	(0.002)	(0.002)
Sample	Labor Force	Part-Time	Labor Force	Part-Time
Observations	4,206,620	916,904	4,206,620	916,904
Panel B: Household Income				
Coefficient	-43.08	435.49**	-158.35	493.98**
(Std. Error)	(101.21)	(223.39)	(450.27)	(251.09)
Sample	Labor Force	Part-Time	IPE	VPE
Observations	4,206,620	916,904	160,059	756,845
Panel C: Usual Hours Worked per Week				
Coefficient	0.088***	0.163***	0.149	0.147***
(Std. Error)	(0.025)	(0.049)	(0.102)	(0.056)
Sample	Labor Force	Part-Time	IPE	VPE
Observations	4,206,620	916,904	160,059	756,845

Notes: This table presents intention-to-treat estimates of the effect of UberX rollout on individual-level outcomes. The coefficient represents the effect of a binary indicator for whether UberX is present in the metropolitan area. Panels A and C use linear probability models; Panel B uses OLS regression. Data come from the Current Population Survey individual monthly files from 2012 to 2019, restricted to the 207 metareas in the sample. Prob(IPE) is the probability an individual is involuntarily part-time employed; Prob(VPE) is the probability an individual is voluntarily part-time employed. Sample columns indicate the population analyzed: Labor Force includes all employed workers, Part-Time includes part-time workers only, IPE includes involuntarily part-time workers, and VPE includes voluntarily part-time workers. Regressions are weighted by the inverse probability of selection (using CPS survey weights). All specifications include controls for demographics, education, occupation, year, and metropolitan area. Household Income was calculated using 2019 dollars. Standard errors, clustered at the metropolitan area level, are shown in parentheses. Observations represent individual-month observations. Significance levels are indicated with asterisks (**, *, *** denote $p < 0.10$, $p < 0.05$, *** $p < 0.01$).

Panel B provides the intention-to-treat effects of UberX on an individual's household incomes broken down by the full labor force sample, the part-time workers sample, and the involuntary and voluntary part-time workers sample. In the entire labor force sample, we find a slight, negative, and statistically insignificant intention-to-treat effect of being exposed to UberX. However, among exclusively part-time workers, we find that exposure to UberX results in an increase of around \$435 in household income for part-time workers living in metareas exposed to UberX compared to part-time workers who live in metareas not exposed to UberX. Among just involuntary part-time workers, we find a negative but statistically insignificant treatment effect on household income. Among just voluntary part-time workers, we find a statistically significant and positive increase in household income of around \$493.

Panel C provides the intention-to-treat effects of UberX on an individual's usual hours worked per week, broken down by the full labor force sample, the part-time workers sample, and the involuntary and voluntary part-time workers sample. Among the full labor force sample, part-time sample, and voluntary part-time sample, we find statistically significant increases in the usual hours worked per week, ranging from 0.09 hours to 0.15 hours, depending on the sample. Among involuntary part-time workers, we find a positive but statistically insignificant effect.

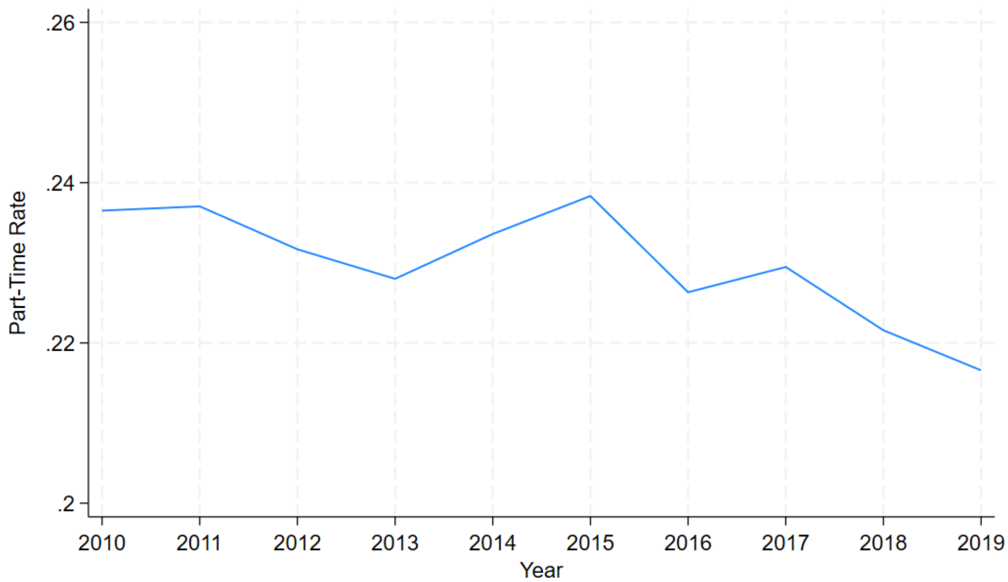
Although the magnitude is small due to the intention-to-treat average effects as opposed to the average treatment effects on treated individuals, who are not directly observed in the CPS, we believe these results are informative of UberX participation at an individual level. If involuntary part-time workers increased their labor supply by turning to UberX, we would expect them to have a lower probability of being involuntary part-time employed, work more hours due to their participation in gig work, and have higher household incomes.

The signs and significance of our results follow the logic of participation in UberX. The probability of being involuntary part-time was lower among the entire labor force and among part-time workers in metareas exposed to UberX compared to individuals in non-exposed metareas. In addition, we have suggestive results that some individuals in exposed metareas might flow into voluntary part-time employment. Household income was higher among part-time workers in areas exposed to UberX being driven by increases in household income from voluntary part-time employees. This results in an increase in the average usual hours worked per week in metareas exposed to

UberX in the entire labor force, especially among part-time workers, driven primarily by an average increase among voluntary part-time employed workers.

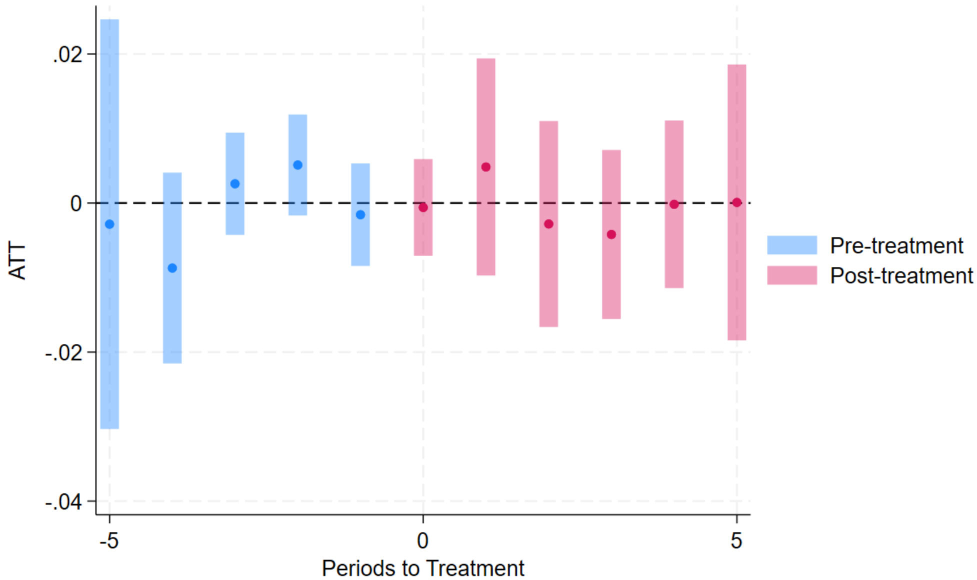
Appendix A.3. Supplementary Figures

Figure A.1: Part-Time as a fraction of Labor Force over Time



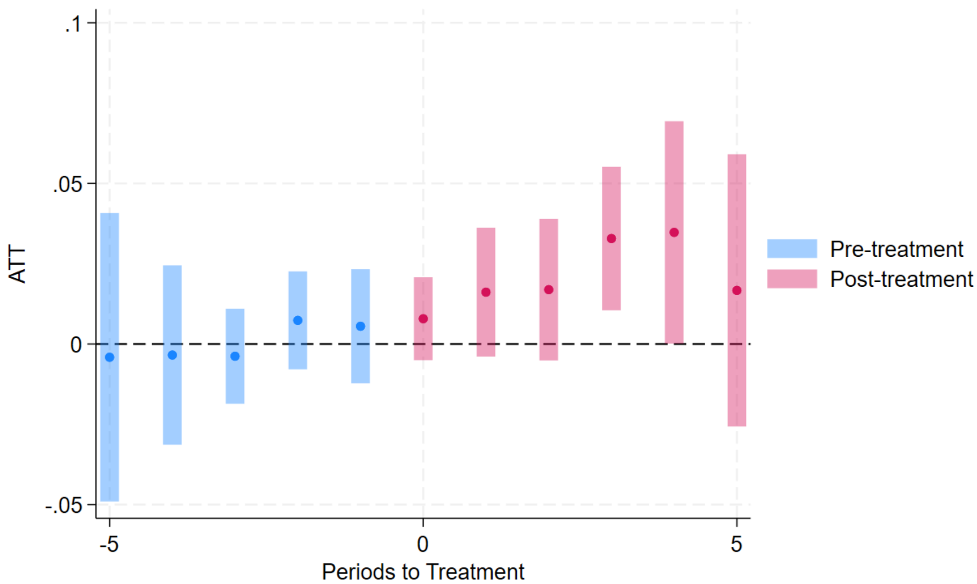
Note: This figure presents the share of the labor force in part-time employment. The involuntarily part-time employed are part-time workers who typically work fewer than 35 hours per week but would prefer a full-time position and are unable to obtain one due to economic conditions. Data come from the Current Population Survey individual monthly files aggregated to the metropolitan area-year level. The figure displays the unweighted average across the 207 metropolitan areas, assigning each metro an equal weight. Individual-level CPS data are aggregated to the metropolitan area-year level using survey weights to ensure each metro-year observation is representative of that area's population. The sample is restricted to metropolitan areas with stable geographic boundaries and data availability from 2010 to 2019.

Figure A.2: Event Study – Voluntary Part-Time Employment as a Share of Labor Force



Note: This figure presents event study estimates of the effect of UberX rollout on voluntary part-time employment as a share of the labor force. Each point represents a treatment effect coefficient relative to UberX entry (period 0), with 95% confidence intervals. Estimates are obtained using the [12] estimator, which employs a varying base period, comparing outcomes in each period to those in the immediately preceding period. This approach produces estimates with confidence intervals for all periods, including pre-treatment periods, allowing for testing of parallel trends. Not-yet-treated metropolitan areas serve as the control group. Data come from the Current Population Survey aggregated to the metropolitan area-year level for 207 metropolitan areas from 2010-2019. Regressions are unweighted, giving each metropolitan area equal weight. Regressions include demographic, age, and education controls. Standard errors are clustered at the metropolitan area level.

Figure A.3: Event Study – Voluntary Part-Time Employment as a Share of Part-Time Employment



Note: This figure presents event study estimates of the effect of UberX rollout on involuntary part-time employment as a share of the part-time employment. Each point represents a treatment effect coefficient relative to UberX entry (period 0), with 95% confidence intervals. Estimates are obtained using the [12] estimator, which employs a varying base period, comparing outcomes in each period to those in the immediately preceding period. This approach produces estimates with confidence intervals for all periods, including pre-treatment periods, allowing for testing of parallel trends. Not-yet-treated metropolitan areas serve as the control group. Data come from the Current Population Survey, aggregated to the metropolitan area-year level, for 207 metropolitan areas from 2010 to 2019. Regressions are unweighted, giving each metropolitan area equal weight. Regressions include demographic, age, and education controls. Standard errors are clustered at the metropolitan area level.

Table A.3: Event Study – Involuntary Part-Time Employment as a Share of Labor Force

	Coefficient	Std. Err.	95% CI
Pre-avg	-0.001	0.001	[-0.003, 0.002]
Post-avg	-0.006**	0.002	[-0.010, -0.002]
T = -5	-0.000	0.007	[-0.014, 0.013]
T = -4	-0.003	0.004	[-0.011, 0.004]
T = -3	0.002	0.002	[-0.003, 0.006]
T = -2	-0.001	0.002	[-0.004, 0.003]
T = -1	-0.001	0.003	[-0.006, 0.003]
T = 0	-0.002	0.002	[-0.005, 0.002]
T = 1	-0.003	0.003	[-0.009, 0.003]
T = 2	-0.005	0.003	[-0.010, 0.001]
T = 3	-0.009***	0.003	[-0.014, -0.003]
T = 4	-0.009*	0.004	[-0.016, -0.001]
T = 5	-0.008*	0.005	[-0.018, -0.001]

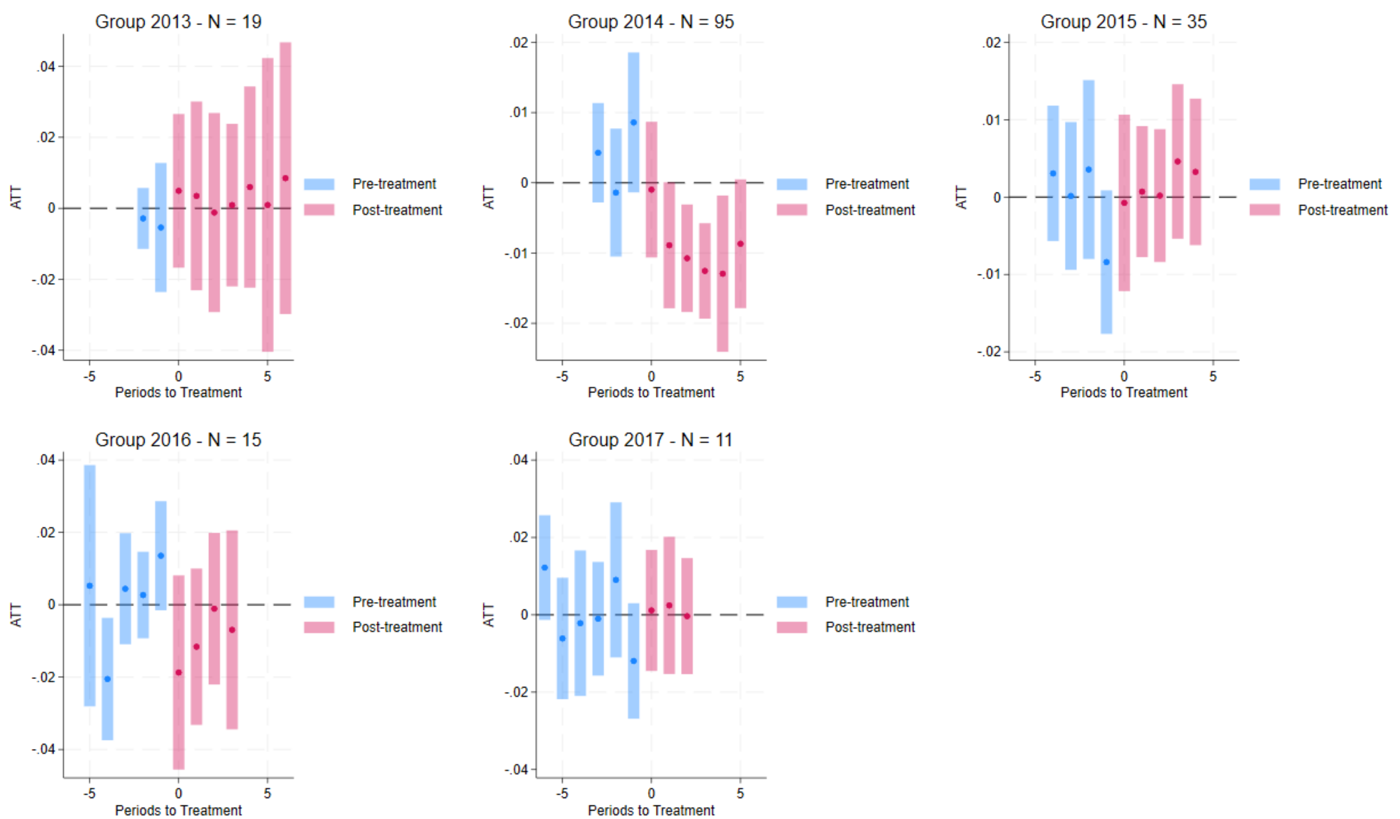
Notes: This table presents event study coefficient estimates corresponding to Figure 4. Each row reports event study coefficients relative to UberX entry (period 0), measuring the effect on involuntary part-time employment as a share of the labor force. 'Pre avg' and 'Post avg' represent the average treatment effects across all pre-treatment and post-treatment periods, respectively. Estimates are obtained using the [12] estimator, which employs a varying base period, comparing outcomes in each period to those in the immediately preceding period. Data come from the Current Population Survey, aggregated to the metropolitan area-year level, for 207 metropolitan areas from 2010 to 2019. Regressions are unweighted, giving each metropolitan area equal weight. Not-yet-treated metropolitan areas serve as the control group. The specification includes demographic, age, and education controls. Standard errors, clustered at the metropolitan area level, are shown in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table A.4: Event Study – Involuntary Part-Time Employment as a Share of Part-Time Employment

	Coefficient	Std. Err.	95% CI
Pre-avg	0.011	0.007	[-0.003, 0.025]
Post-avg	-0.019	0.012	[-0.043, 0.005]
T = -5	0.017	0.024	[-0.031, 0.065]
T = -4	0.010	0.016	[-0.022, 0.041]
T = -3	0.017*	0.009	[-0.000, 0.035]
T = -2	0.005	0.014	[-0.021, 0.032]
T = -1	0.007	0.012	[-0.016, 0.030]
T = 0	-0.002	0.019	[-0.038, 0.035]
T = 1	-0.014	0.015	[-0.043, 0.016]
T = 2	-0.020*	0.010	[-0.040, 0.000]
T = 3	-0.025**	0.012	[-0.049, -0.002]
T = 4	-0.035**	0.018	[-0.069, -0.000]
T = 5	-0.017	0.022	[-0.059, 0.026]

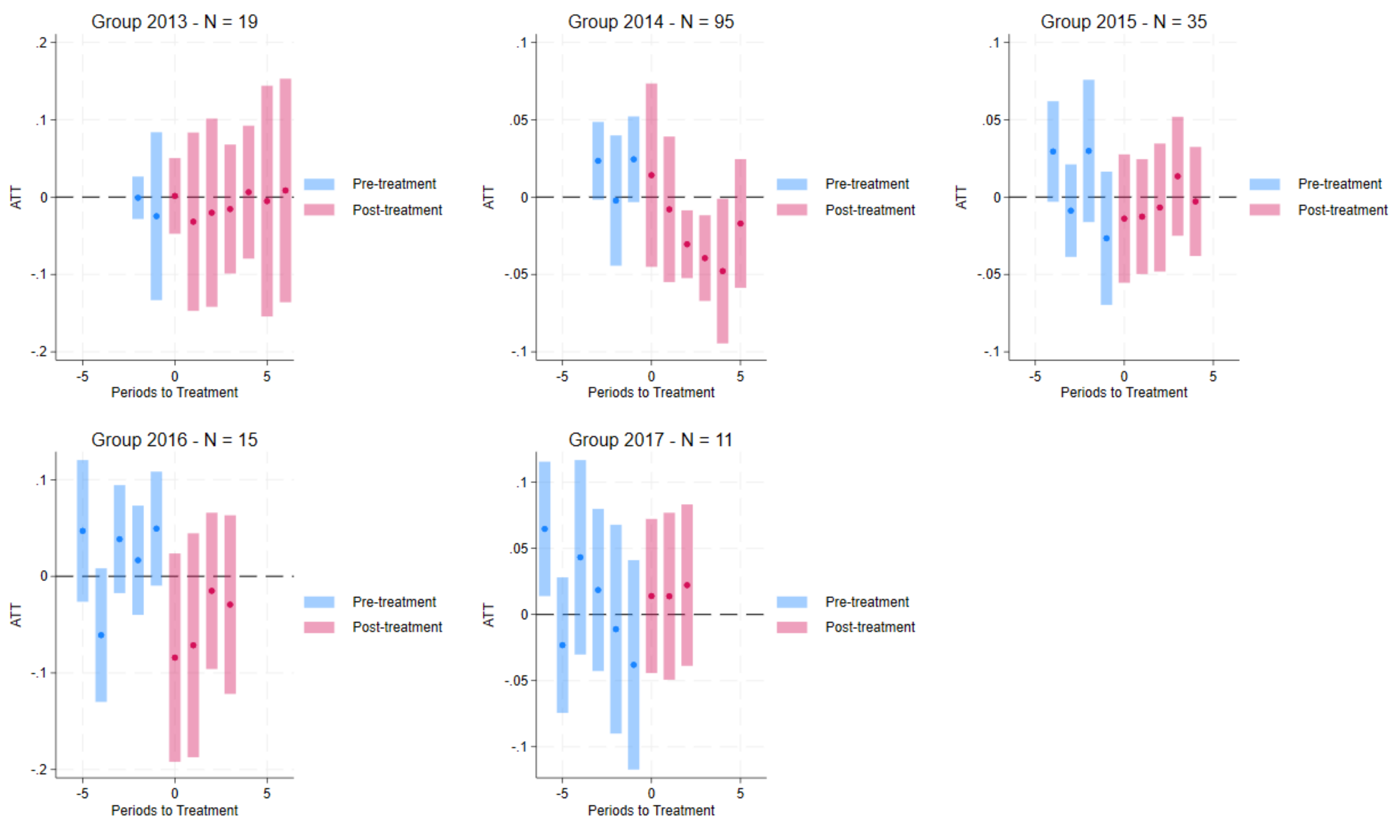
Notes: This table presents event study coefficient estimates corresponding to Figure 5. Each row reports event study coefficients relative to UberX entry (period 0), measuring the effect on involuntary part-time employment as a share of the part-time employment. 'Pre avg' and 'Post avg' represent the average treatment effects across all pre-treatment and post-treatment periods, respectively. Estimates are obtained using the [12] estimator, which employs a varying base period, comparing outcomes in each period to those in the immediately preceding period. Data come from the Current Population Survey, aggregated to the metropolitan area-year level, for 207 metropolitan areas from 2010 to 2019. Regressions are unweighted, giving each metropolitan area equal weight. Not-yet-treated metropolitan areas serve as the control group. The specification includes demographic, age, and education controls. Standard errors, clustered at the metropolitan area level, are shown in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% levels, respectively.

Figure A.4: Event Study – Involuntary Part-time Employment as a Fraction of Total Labor Force – ATT by Group



Notes: This figure presents event study estimates of the effect of UberX rollout on involuntary part-time employment as a share of the labor force, separately by treatment cohort. Each point represents a treatment effect coefficient relative to UberX entry (period 0), with 95% confidence intervals. Treatment cohorts indicate the year UberX first entered the metropolitan area. Estimates are obtained using the [12] estimator, which employs a varying base period, comparing outcomes in each period to those in the immediately preceding period. Not-yet-treated metropolitan areas serve as the control group. Data come from the Current Population Survey, aggregated to the metropolitan area-year level, for 207 metropolitan areas from 2010 to 2019. Regressions are unweighted, giving each metropolitan area equal weight. Regressions include demographic, age, and education controls. Standard errors are clustered at the metropolitan area level.

Figure A.5: Event Study – Involuntary Part-time Employment as a Fraction of Total Part-time Employment – ATT by Group



Notes: This figure presents event study estimates of the effect of UberX rollout on involuntary part-time employment as a share of part-time employment, separately by treatment cohort. Each point represents a treatment effect coefficient relative to the UberX entry (period 0), with 95% confidence intervals. Treatment cohorts indicate the year UberX first entered the metropolitan area. Estimates are obtained using the Callaway and Sant'Anna (2021) estimator with a varying base period, which compares outcomes in each period to the immediately preceding period. Not-yet-treated metropolitan areas serve as the control group. Data come from the Current Population Survey aggregated to the metropolitan area-year level for 207 metropolitan areas from 2010-2019. Regressions are unweighted, giving each metropolitan area equal weight. Regressions include demographic, age, and education controls. Standard errors are clustered at the metropolitan area level.

Appendix B. Additional Robustness

Appendix B.1. Alternative Event Studies

Table B.1: Event Study Coefficients (IPE as a Share of the Labor Force) - Never Treated

	(1) No Controls	(2) Age	(3) Age + Demo	(4) Full
Pre-Avg.	0.000	0.001	0.000	-0.001
	(0.001)	(0.001)	(0.001)	(0.001)
Post-Avg.	-0.004	-0.003	-0.004	-0.006**
	(0.003)	(0.003)	(0.002)	(0.002)
T = -5	0.004	0.005	0.004	-0.000
	(0.005)	(0.005)	(0.005)	(0.007)
T = -4	-0.002	-0.001	-0.002	-0.003
	(0.003)	(0.003)	(0.003)	(0.004)
T = -3	0.001	0.002	0.002	0.002
	(0.002)	(0.002)	(0.002)	(0.002)
T = -2	-0.001	-0.001	-0.001	-0.001
	(0.002)	(0.002)	(0.002)	(0.002)
T = -1	0.001	0.000	0.001	-0.001
	(0.002)	(0.002)	(0.002)	(0.003)
T = 0	-0.001	0.000	0.000	-0.002
	(0.002)	(0.002)	(0.002)	(0.002)
T = +1	-0.003	-0.002	-0.003*	-0.003
	(0.002)	(0.002)	(0.002)	(0.003)
T = +2	-0.004	-0.004	-0.006*	-0.005*
	(0.003)	(0.003)	(0.003)	(0.003)
T = +3	-0.005	-0.006**	-0.008**	-0.009***
	(0.003)	(0.003)	(0.003)	(0.003)
T = +4	-0.004	-0.004	-0.005	-0.009*
	(0.004)	(0.004)	(0.004)	(0.004)
T = +5	-0.005	-0.004	-0.004	-0.008*
	(0.005)	(0.004)	(0.004)	(0.005)

Notes: This table presents event study coefficient estimates using never-treated metropolitan areas as the control group, as a robustness check on the main specification. Each row reports event study coefficients relative to UberX entry (period 0), measuring the effect on involuntary part-time employment as a share of the labor force. Column (1) includes no controls; (2) includes age composition controls; (3) adds demographic controls (sex, race); (4) adds education controls. Pre-Avg. and Post-Avg. report average effects across pre-treatment and post-treatment periods, respectively. Estimates are obtained using the [12] estimator, which employs a varying base period, comparing outcomes in each period to those in the immediately preceding period. Data come from the Current Population Survey, aggregated to the metropolitan area-year level, for 207 metropolitan areas from 2010 to 2019. Regressions are unweighted, giving each metropolitan area equal weight. Never-treated metropolitan areas (those not receiving UberX by 2019) serve as the control group. Standard errors, clustered at the metropolitan area level, are shown in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table B.2: Event Study Coefficients (IPE as a Share of Part-Time Employment) - Never Treated

	(1) No Controls	(2) Age	(3) Age + Demo	(4) Full
Pre-Avg.	0.001	0.004	0.005	-0.000
	(0.004)	(0.005)	(0.005)	(0.004)
Post-Avg.	-0.017	-0.014	-0.017	-0.021
	(0.009)	(0.008)	(0.008)	(0.009)
T = -5	0.018	0.022	0.020	0.004
	(0.019)	(0.019)	(0.019)	(0.023)
T = -4	-0.004	0.005	0.006	0.003
	(0.012)	(0.012)	(0.012)	(0.014)
T = -3	-0.005	-0.003	0.001	0.004
	(0.007)	(0.008)	(0.008)	(0.008)
T = -2	-0.005	-0.006	-0.007	-0.007
	(0.006)	(0.007)	(0.008)	(0.008)
T = -1	0.001	0.003	0.004	-0.006
	(0.006)	(0.007)	(0.007)	(0.009)
T = 0	-0.002	0.000	-0.002	-0.008
	(0.006)	(0.007)	(0.007)	(0.007)
T = +1	-0.010	0.005	-0.008	-0.016
	(0.008)	(0.011)	(0.011)	(0.010)
T = +2	-0.019	-0.016	-0.023	-0.017
	(0.011)	(0.010)	(0.011)	(0.011)
T = +3	-0.027	-0.024	-0.031	-0.033
	(0.011)	(0.012)	(0.012)	(0.011)
T = +4	-0.025	-0.024	-0.028	-0.035
	(0.014)	(0.015)	(0.016)	(0.018)
T = +5	-0.023	-0.014	-0.012	-0.017
	(0.016)	(0.015)	(0.017)	(0.022)

Notes: This table presents event study coefficient estimates using never-treated metropolitan areas as the control group, as a robustness check on the main specification. Each row reports event study coefficients relative to UberX entry (period 0), measuring the effect on involuntary part-time employment as a share of the part-time employment. Column (1) includes no controls; (2) includes age composition controls; (3) adds demographic controls (sex, race); (4) adds education controls. Pre-Avg. and Post-Avg. report average effects across pre-treatment and post-treatment periods, respectively. Estimates are obtained using the [12] estimator, which employs a varying base period, comparing outcomes in each period to those in the immediately preceding period. Data come from the Current Population Survey, aggregated to the metropolitan area-year level, for 207 metropolitan areas from 2010 to 2019. Regressions are unweighted, giving each metropolitan area equal weight. Never-treated metropolitan areas (those not receiving UberX by 2019) serve as the control group. Standard errors, clustered at the metropolitan area level, are shown in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% levels, respectively.

Appendix B.2. Main Results Robustness

Table B.3: Static Average Treatment Effects on Involuntary Part-Time Employment - Never Treated Comp.

Panel A: Involuntary Part-Time as a Share of the Labor Force

	(1) No Controls	(2) Age	(3) Age + Demo	(4) Full
ATT	-0.003	-0.003	-0.004*	-0.006**
	(0.003)	(0.002)	(0.002)	(0.002)
Observations	2,070	2,070	2,070	2,070
2012 Mean	0.05	0.05	0.05	0.05

Panel B: Involuntary Part-Time as a Share of Part-Time Employment

	(1) No Controls	(2) Age	(3) Age + Demo	(4) Full
ATT	-0.018**	-0.013	-0.018*	-0.018
	(0.009)	(0.009)	(0.009)	(0.012)
Observations	2,070	2,070	2,070	2,070
2012 Mean	0.23	0.23	0.23	0.23

Notes: This table presents static average treatment effects (ATT) of UberX entry on involuntary part-time employment using never-treated metropolitan areas as the control group, as a robustness check on the main specification. Panel A uses IPE as a share of the labor force as the dependent variable; Panel B uses IPE as a share of part-time employment. Estimates are obtained using the [12] estimator. Column (1) includes no controls; (2) includes age composition controls; (3) adds demographic controls (sex, race); (4) adds education controls. Data come from the Current Population Survey, aggregated to the metropolitan area-year level, for 207 metropolitan areas from 2010 to 2019. Regressions are unweighted, giving each metropolitan area equal weight. The 2012 Mean shows the pre-treatment baseline mean of the dependent variable. Never-treated metropolitan areas (those not receiving UberX by 2019) serve as the control group. Standard errors, clustered at the metropolitan area level, are shown in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% levels, respectively. Observations represent metropolitan area-year combinations.